CHAPTER
06
STAGE 6 OF 15

Deposition and Etch

Deposition and etch tools build a chip layer by layer. They are among the chipmaking tools where Chinese toolmakers gained the most share from 2019 to 2024.

IN PLAIN TERMS

Deposition and etch build the chip itself, one layer at a time. Deposition lays down a thin film, sometimes only a few atoms thick. Lithography prints a pattern on it. Etch then eats away everything the pattern does not protect, and the next film goes down on top. Making one chip takes 1,000 to 1,500 steps, the Semiconductor Industry Association estimated in 2025[1]. Some of the holes the etch cuts in memory chips are more than 50 times deeper than they are wide, like a finger-wide shaft about a meter deep[2]. The film has to coat them right to the bottom. The etch has to eat one material and leave the one beside it untouched. Applied Materials and Lam Research in America, Tokyo Electron in Japan and ASM International in the Netherlands make most of these machines. Chinese toolmakers have gained share in both kinds of tool since 2019.

IN SHORT

American companies held 60.3 percent of the world market for deposition tools in 2025, Japanese companies 14.0 percent and China-based companies 9.8 percent, by CSET's count from TechInsights data[3]. In etch and cleaning tools that year, American companies held 52.6 percent, Japanese companies 32.6 percent and China-based companies 10.1 percent[4].

Atomic layer deposition tools coat a hole's walls and floor as evenly as its rim[5]. China-based companies held only 2.6 percent of the world market for them in 2025, by CSET's count from TechInsights data[6]. The December 2024 US export rule targets tools for advanced atomic layer deposition and for coating holes more than 200 times deeper than they are wide[7].

1 Testimony of David Isaacs, Semiconductor Industry Association, hearing on... U.S. Senate Committee on Finance, 14 May 2025, p. 2. finance.senate.gov

2 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 17. govinfo.gov

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

4 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

5 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 16. govinfo.gov

6 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

7 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 33. govinfo.gov

CHOKEPOINT CARD	HIGH CONCENTRATION	MODERATELY HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
High	Moderate In 2025 Lam Research held 41.1 percent of dry etch tools, Tokyo Electron 23.4 percent and Applied Materials 17.4 percent, by CSET's count from TechInsights data[8].	$26.2B deposition tool market in 2025, by CSET's count from TechInsights data[3]. Dry etch tools were a $21.9B market in 2025[8]

WHO LEADS	US **Applied Materials** 82.4% of physical vapor deposition tools in 2025, by CSET's count from TechInsights data[9] US **Lam Research** 51.2% of conductor etch tools in 2025[10] JP **Tokyo Electron** 55.9% of dielectric etch tools in 2025[11] NL **ASM International** 54.5% of atomic layer deposition tools in 2025[6] CN **Naura** 14.7% of physical vapor deposition tools in 2025[9]
WHERE IT IS MADE	US **United States** Applied Materials, Lam Research JP **Japan** Tokyo Electron, Kokusai Electric NL **Netherlands** ASM International, atomic layer deposition KR **South Korea** Large equipment buyer CN **China** Naura, AMEC and Piotech, in the largest equipment market ($49.3B in 2025)
WHY SUBSTITUTION IS POSSIBLE	Atomic layer deposition lays down a film a fraction of a layer of atoms at a time[5]. At least five makers sell it[6].

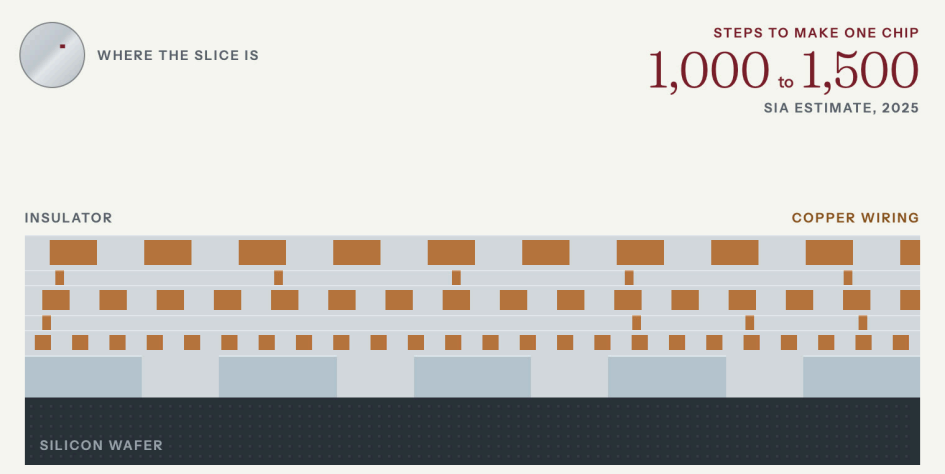
WHERE CHINA STANDS	WHERE THE US STANDS
Chinese toolmakers held 12.4 percent of dry etch in 2025, according to CSET's analysis of TechInsights data[8]. CSET's earlier analysis of TechInsights data had them under 1 percent of atomic layer deposition in 2024[12].	A December 2024 US rule lists some dry etch, atomic layer deposition and deep-hole deposition tools as controlled exports[13]. Its foreign direct product rule reaches some tools built outside the United States[14].

Applied Materials and Lam Research are the two largest deposition and etch suppliers. Applied Materials booked $28.368 billion of revenue in fiscal 2025[15]. Lam Research booked $23.233 billion in fiscal 2026[16].

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
5 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 16. govinfo.gov
6 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
8 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
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11 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
12 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu
13 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 32. govinfo.gov
14 Commerce Strengthens Export Controls to Restrict China's Capability to Produce... U.S. Bureau of Industry and Security, 2 December 2024. bis.gov
15 Revenue from Contract with Customer, Excluding Assessed Tax... U.S. Securities and Exchange Commission, accessed 9 September 2026. data.sec.gov
16 LAM RESEARCH CORP, Form 8-K current report for the period ended 2026-07-29 (8-K). U.S. Securities and Exchange Commission, 29 July 2026. sec.gov

How it works

One round of deposition and etch



1. **The wafer.** A slice through one spot on a wafer, magnified until its layers show.
2. **Deposit.** A thin film is laid over the whole wafer.
3. **Coat.** A light-sensitive coating, the resist, is spun on top.
4. **Expose.** Light shines through a stencil and changes the resist wherever it lands.
5. **Develop.** A wash removes the resist the light reached, leaving the stencil's pattern.
6. **Etch.** A plasma cuts straight down through the film wherever the resist is gone.
7. **Strip.** The leftover resist is removed. The pattern now sits in the film.
8. **Repeat.** Insulator fills the gaps and the next layer goes on. Making one chip takes 1,000 to 1,500 steps, the Semiconductor Industry Association estimated in 2025.

Simplified cross-section. The layers are drawn much thicker than they are.

Most etch steps, and many deposition steps, happen in a vacuum chamber that holds one wafer at a time. A pump empties the chamber of air, and a measured flow of gas comes in. A radio-frequency field, an electric field that reverses millions of times a second, turns the gas into a plasma. The field splits the gas into charged ions and into neutral fragments called radicals, which are more reactive than whole molecules[17]. Batch furnaces such as Tokyo Electron's TELINDY PLUS instead heat as many as 125 wafers at once, usually without plasma, for oxidation, low-pressure chemical vapor deposition and some atomic layer deposition[18].

In most deposition steps, the gas carries the material the film will be made of. The molecules, or the radicals a plasma splits from them, land on the wafer and stick. The film builds up a few atoms at a time.

To coat the floor of a deep hole as thickly as its rim, the machine can work in cycles, a method called atomic layer deposition. First one gas settles in a single layer on every surface, and the pump removes the rest. Then a second gas reacts with that layer and nothing else, and the pump empties the chamber again. Each cycle adds a fraction of a layer of atoms wherever the

17 Etch Essentials: The Building Blocks of AI Era Microchips (Semi 101). Lam Research, 12 June 2024. newsroom.lamresearch.com
18 TELINDY™ Series. Tokyo Electron, accessed 9 September 2026. tel.com

gas reached, so the film grows as evenly on a hole's walls and floor as on its rim[5].

For etch, the gas attacks one material and leaves the one beside it. Wherever the photoresist covers the wafer, it shields the film beneath. Wherever it does not, the radicals react with the surface and turn it into a gas, which the pump carries away. A second field pulls the ions straight down at the wafer, so the etch cuts vertically and a narrow hole keeps its width as it deepens[17]. When the etch reaches the material underneath, which the gas was chosen not to touch, it slows almost to a stop, and the tool ends the step.

The US export controls describe each deposition or etch tool by the process it is designed to run[19].

Accenture and the Global Semiconductor Alliance estimated in 2020 that a part-finished chip can cross international borders 70 or more times on its way to the end customer[1]. The Semiconductor Industry Association repeated the figure in 2025[1].

Variants and trade-offs

Deposition families

The market sizes and leading shares below are 2025 figures from Georgetown University's Center for Security and Emerging Technology (CSET), based on data from the research firm TechInsights. CSET puts the deposition tool market at $26.2 billion[3]. It puts the dry etch market at $21.9 billion[8].

- **Physical vapor deposition** (PVD), which knocks atoms off a solid target and lands them on the wafer, was a $4.9 billion market in 2025[9]. Applied Materials held 82.4 percent of it and China's Naura 14.7 percent[9].
- **Chemical vapor deposition** (CVD), which reacts gases at the wafer surface to grow a film, was an $11.2 billion market in 2025, led by Applied Materials at 47 percent[20]. The plasma-enhanced version (PECVD) runs cooler and made up most of it, $7.5 billion[21].
- **Atomic layer deposition** (ALD) lays down less than one atomic layer at a time. Each step stops once every surface is covered, so it coats vertical walls as evenly as horizontal floors. US regulators say that even coating makes it important for building 3D structures, such as 3D DRAM, 3D NAND, and the FinFET and gate-all-around transistors in logic chips[5]. ASM International leads it with 54.5 percent of a $4.1 billion market in 2025[6].
- **Epitaxy** grows layers that line up with the crystal beneath them, using high-temperature CVD tools[22]. **Electroplating** deposits the copper used for

1 Testimony of David Isaacs, Semiconductor Industry Association, hearing on... U.S. Senate Committee on Finance, 14 May 2025, p. 2. finance.senate.gov
3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
5 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 16. govinfo.gov
6 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
8 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
9 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
17 Etch Essentials: The Building Blocks of AI Era Microchips (Semi 101). Lam Research, 12 June 2024. newsroom.lamresearch.com
19 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 28. govinfo.gov
20 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
21 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
22 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

wiring across the wafer[23].

Etch families

- **Conductor etch**, which cuts gates, metals and silicon, was a $12.2 billion market in 2025, led by Lam at 51.2 percent, by CSET's count from TechInsights data[10].
- **Dielectric etch**, which cuts the insulating oxides and nitrides between them, was an $8.7 billion market in 2025, led by Tokyo Electron at 55.9 percent, by CSET's count from TechInsights data[11].
- **High-aspect-ratio etch** cuts the deep, narrow channels that let 3D NAND stack layers of memory cells[24]. Lam says slight departures from a channel's intended shape can hurt yield[24]. Lam runs this etch well below 0 degrees Celsius[24]. It says its Cryo 3.0 technology can etch channels up to 10 microns deep whose width varies from top to bottom by less than 0.1 percent of their depth[24].

AI hardware needs sideways cuts, deeper holes and a second stack of metal wiring on the back of the wafer.

- **Gate-all-around** transistors need highly selective isotropic etch, which cuts sideways[25]. The US Bureau of Industry and Security (BIS) controls isotropic dry etch tools that remove silicon-germanium at least 100 times faster than silicon, a selectivity of 100:1 or more[19].
- **HBM** (high-bandwidth memory) needs holes etched straight through the silicon, called through-silicon vias (TSVs), so chips can be stacked and wired[26]. BIS controls the tools that etch these holes at aspect ratios (depth to width) of 10:1 or more and at rates above 7 microns a minute[13]. The control covers only tools that also keep the holes' depth even to within 2 percent across the wafer[13].
- **Backside power delivery** moves the wires that carry power to the back of the wafer[27]. In July 2024 Applied Materials said it expected backside power to add $1 billion to the market for its wiring tools, for every 100,000 wafers a month of new fab capacity[28].

10 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

11 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech

13 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for... U.S. Government Publishing Office, 5 December 2024, p. 32. govinfo.gov

19 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 28. govinfo.gov

23 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology, 21 January 2021, p. 39. cset.georgetown.edu

24 Lam Research Introduces Lam Cryo™ 3.0 Cryogenic Etch Technology to Accelerate... Lam Research, 31 July 2024. newsroom.lamresearch.com

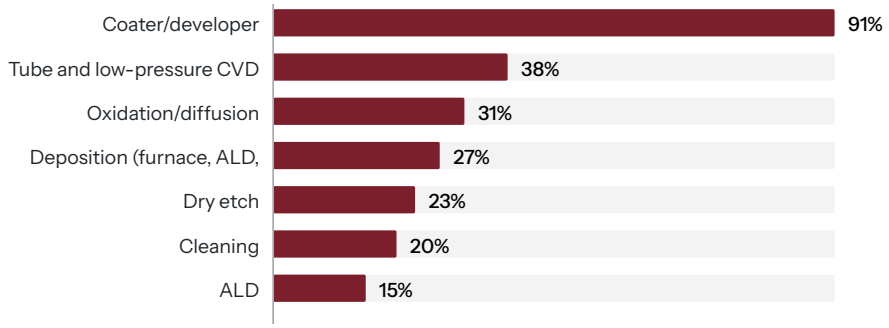
25 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 15. govinfo.gov

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27 PowerVia Test Shows Industry-Leading Performance. Intel, 5 June 2023. download.intel.com

28 Applied Materials Unveils Chip Wiring Innovations for More Energy-Efficient... GlobeNewswire (release by Applied Materials), 8 July 2024. globenewswire.com

Tokyo Electron world market share by tool type, 2025 %



Source: Gartner, as reported in Tokyo Electron's FY2026 Q4 results presentation

Who makes it

The industry association SEMI put world sales of chipmaking equipment at $135.1 billion in 2025, up 15 percent from $117.1 billion[30]. Sales of wafer processing equipment, the category that includes deposition and etch tools, rose 12 percent[30].

No one firm dominates deposition and etch the way ASML dominates lithography. In dry etch, CSET's count for 2025 from TechInsights data puts Lam Research at 41.1 percent, Tokyo Electron at 23.4 percent and Applied Materials at 17.4 percent[8]. Tokyo Electron's own results presentation, citing Gartner, gives it 23 percent of dry etch, 27 percent of deposition systems (furnace, ALD and low-pressure CVD tools) and 15 percent of ALD in 2025[29]. The same presentation, again citing Gartner, gives it 91 percent of coater/developers in 2025[29]. These machines coat wafers with photoresist and develop the pattern after lithography.

ASM International, the atomic layer deposition specialist, reached a record EUR 3.2 billion of revenue in 2025 as customers built 2 nm gate-all-around capacity[31]. Molybdenum ALD, a new use of its tools that deposits the metal molybdenum, entered volume production that year[31].

What the world spent on chipmaking equipment $B



Source: SEMI, Worldwide Semiconductor Equipment Market Statistics

China and the controls

China is the tool vendors' largest market, buying $49.3 billion of equipment in 2025, half a percent less than in 2024[30]. Taiwan's spending rose 90 percent to $31.5 billion[30]. South Korea's rose 26 percent to $25.8 billion[30]. China still takes a large share of every major vendor's sales.

8 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
29 FY2026 (April 2025-March 2026) Financial Announcement, transcript. Tokyo Electron, 30 April 2026, p. 28. tel.com
30 SEMI Reports Global Semiconductor Equipment Billings Reached $135 Billion in... SEMI, 7 April 2026. semi.org
31 ASM reports Q4 and full-year 2025 results. ASM International, 3 March 2026, p. 2. asm.com

- **Tokyo Electron**: 34.1 percent of sales in the year to March 2026, down from 41.7 percent[32]. It was 30.4 percent in the quarter to June 2026[33].
- **Lam Research**: about 34 percent of revenue in the year to June 2026, the same as a year earlier[34].
- **Applied Materials**: 28 percent in the third fiscal quarter of 2026, down from 35 percent a year earlier[35].
- **ASM International**: more than 30 percent of 2025 revenue[31].

Between 2019 and 2024, by CSET's analysis of TechInsights data, China-based suppliers went from 2 to 7 percent of deposition and from under 3 to almost 9 percent of etch and clean[12]. In ALD they gained no share and held less than 1 percent in 2024[12].

CSET's newer analysis of data that TechInsights published in July 2026 puts them at 9.8 percent of deposition in 2025[3]. They held 10.1 percent of etch and clean that year[4]. In dry etch alone their share was 12.4 percent[8]. In ALD they held 2.6 percent in 2025, most of it from Piotech[6].

The Chinese toolmaker AMEC alone held 12.4 percent of dielectric etch in 2025, by CSET's count from TechInsights data[11]. Its revenue was RMB 12.385 billion that year, up about 36.62 percent, and it spent RMB 3.744 billion on research and development, 30.2 percent of sales[36].

The United States tightened its export controls at least five times between 2022 and 2025.

- **October 2022.** The 7 October interim final rule required a license for certain chipmaking equipment, and for some activities by US persons that support the making of certain chips in China[37]. It also extended US license rules to more foreign-made items bound for 28 Chinese entities on the Entity List, the US list of foreign firms that exporters need a license to supply[37].
- **October 2023.** The October rule, effective 17 November, rewrote the equipment entries[38]. It added sideways (isotropic) and vertical (anisotropic) dry etch (ECCN 3B001.c.1), spatial ALD (3B001.d.9), low-fluorine tungsten ALD and CVD (3B001.d.10) and carbon hard mask PECVD (3B001.d.5)[5].

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO), accessed 9 September 2026. chipexplorer.eto.tech
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12 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu
31 ASM reports Q4 and full-year 2025 results. ASM International, 3 March 2026, p. 2. asm.com
32 FY2026 (April 2025-March 2026) Financial Announcement, transcript. Tokyo Electron, 30 April 2026, p. 27. tel.com
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37 Implementation of Additional Export Controls: Certain Advanced Computing and... U.S. Government Publishing Office, 13 October 2022, p. 1. govinfo.gov
38 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office, 25 October 2023, p. 1. govinfo.gov

- **December 2024.** BIS controlled 24 more types of equipment and three types of software, and added 140 entities to the Entity List[14]. The new equipment entries include TSV etch, and deposition into features with aspect ratios above 200:1[39]. BIS also wrote a foreign direct product rule that reaches some tools built outside the United States[14]. The rule largely exempts foreign-made tools exported by firms located in Japan, the Netherlands and the 31 other countries it lists, whose governments can control these tools themselves[40]. Japan added advanced chipmaking equipment to its control list on 23 July 2023[41].
- **September 2025.** BIS removed the Chinese units of Samsung and SK hynix, and Intel Semiconductor (Dalian), from the Validated End-User program, effective 31 December 2025[42]. From that date, US tools bound for them need a license[42]. BIS estimated that the change would add 1,000 license applications a year[42].
- **September 2025.** The Affiliates Rule extended Entity List restrictions to any company at least 50 percent owned by a listed entity[43]. BIS then suspended it until 9 November 2026[44].

The rules apply to the toolmakers as well as to their customers. In fiscal 2026 Applied Materials took a $253 million charge to settle an export controls compliance matter with BIS[35].

Of the tools in the chart below, China-based suppliers have their largest share in dry stripping, which clears leftover photoresist off the wafer and was a $314.6 million market in 2025, by CSET's count from TechInsights data[45]. Kokusai held 35.7 percent of it, Mattson Technology 22.6 percent and Naura 8.3 percent[45]. Mattson is a US-based firm that Beijing E-Town bought in 2016[12].

12 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology, 14 July 2025. cset.georgetown.edu

14 Commerce Strengthens Export Controls to Restrict China's Capability to Produce... U.S. Bureau of Industry and Security, 2 December 2024. bis.gov

35 Applied Materials Announces Third Quarter 2026 Results. GlobeNewswire (release by Applied Materials), 13 August 2026. globenewswire.com

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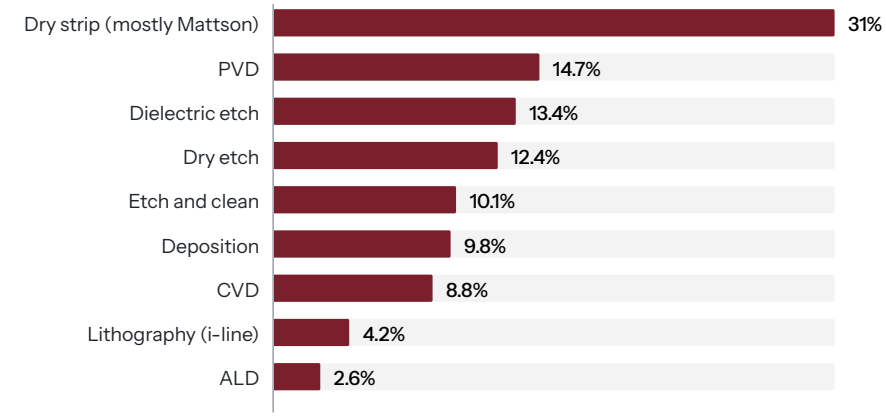
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China-based suppliers' share of global tool segments, 2025 %



Source: CSET analysis of TechInsights data (published July 2026), ETO Chip Explorer

The chokepoint

Deposition and etch tools are a chokepoint, but a weaker one than lithography. Most deposition and etch steps have three or four credible vendors. Atomic layer deposition has at least five[6]. The December 2024 rule targets tools for advanced atomic layer deposition and for coating holes more than 200 times deeper than they are wide[7].

FOUR THINGS TO REMEMBER

- Deposition lays down thin films and etch cuts them into patterns. Making one chip takes 1,000 to 1,500 steps, by a 2025 industry estimate.
- Four American, Japanese and Dutch firms make most of these tools, and none dominates.
- Chinese makers held 12.4 percent of dry etch and 9.8 percent of deposition in 2025, on CSET's count.
- Chinese makers held 2.6 percent of atomic layer deposition in 2025, most of it from Piotech.

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Made by **Yashvardhan Sharma** with Claude. Corrections and questions: contact@chipsupplychain.org.

UPDATED 20 OCTOBER 2026

A company filings, governments, standards bodies, universities and research institutions; B SemiAnalysis, Reuters, the Financial Times and Nikkei Asia.

HOW TO CITE THIS CHAPTER

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Sharma, Yashvardhan. “Deposition and Etch.” Chip Supply Chain: How AI Chips Are Made.
Edition of 20 October 2026. https://chipsupplychain.org/deposition-and-etch/
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@misc{chipsupplychain2026depositionandetch,
  title        = {Deposition and Etch},
  author       = {Sharma, Yashvardhan},
  howpublished = {Chip Supply Chain: How AI Chips Are Made},
  year         = {2026},
  month        = {October},
  note         = {Edition of 20 October 2026},
  url          = {https://chipsupplychain.org/deposition-and-etch/}
}
```